

1 Constraint Model

We adopt the natural mixed-integer formulation of capacitated facility location with vehicle workload, and then relax all integrality so that every decision variable becomes continuous on a real interval. The resulting LP is what `src/lpinstance.py` gives to OR-Tools.

We use an aggregated form: v_f counts vehicles at facility f as a single continuous quantity, rather than indexing each vehicle slot separately. This keeps the model compact (one variable per facility for vehicles, one per customer-facility pair for assignment) at the cost of some LP tightness: distance is constrained in aggregate, not per truck. The trade-off is acceptable because LP only needs to certify a lower bound, not implement a plan.

2 Decision Variables

Let $C = \{1, \dots, \text{numCustomers}\}$ and $F = \{1, \dots, \text{numFacilities}\}$ denote the customer and facility index sets, and let $V := \text{numMaxVehiclePerFacility}$ denote the per-facility upper bound on the vehicle count, which `load_from_file` sets to $|C|$. (Per the handout: in the worst case each allocated customer can need its own vehicle.)

- $y_f \in [0, 1]$, one per facility $f \in F$: the fraction to which facility f is opened. In IP, this is binary. In LP, it can be any real value in $[0, 1]$.
- $x_{c,f} \in [0, 1]$, one per pair $(c, f) \in C \times F$: the fraction of customer c 's demand that is served from facility f . In IP, this is binary (each customer is assigned to exactly one facility). However, LP allows split assignments.
- $v_f \in [0, V]$, one per facility $f \in F$: the continuous number of vehicles deployed at facility f . In IP, this is an integer value.

3 Constraints

1. Each customer is fully served across some combination of facilities:

$$\sum_{f \in F} x_{c,f} = 1 \quad \forall c \in C.$$

In IP, $x_{c,f} \in \{0, 1\}$ would force a customer to be served by exactly one open facility; the LP lets $x_{c,f}$ take any value in $[0, 1]$, permitting convex combinations across multiple facilities.

2. A customer may only draw service from a facility to the extent that facility is open:

$$x_{c,f} \leq y_f \quad \forall c \in C, f \in F.$$

(In IP, it would have to be $x = 1 \Rightarrow y = 1$.)

Project – III

3. The demand served from a facility f may not exceed its capacity, scaled by its openness. (A half-open facility carries only half the capacity.)

$$\sum_{c \in C} d_c x_{c,f} \leq q_f y_f \quad \forall f \in F.$$

4. Vehicle workload: Total round-trip distance dispatched from a facility f cannot exceed the workload that v_f vehicles can absorb:

$$\sum_{c \in C} \delta_{c,f} x_{c,f} \leq T v_f \quad \forall f \in F.$$

We have this instead of a per-vehicle limit. With v_f being continuous, we can split distance across fractional vehicles.

5. A closed facility cannot deploy vehicles, and we cannot deploy more than V :

$$v_f \leq V \cdot y_f \quad \forall f \in F.$$

Objective. Minimise the sum of opening, service, and truck-usage costs:

$$\min \sum_{f \in F} o_f y_f + \sum_{c \in C} \sum_{f \in F} a_{c,f} x_{c,f} + u \sum_{f \in F} v_f.$$

4 Effort

Approximately 20 hours, including modeling, debugging, and writing this report.